

Andrey Zaytsev

📍 Lisbon, Portugal · Open to relocation across Europe

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Senior Product & Application Security Engineer with 9+ years across Google and Jump Trading, including 5.5 years securing Chrome. Specialised in threat modelling, secure design/code review, vulnerability remediation, and secure SDLC.

Experience

Independent Product Engineer

Jul 2025 - Present

- Designed, built, and operate solange.top, a production React/Tailwind product hosted on Vercel with Cal.com and Stripe integrations; own requirements, implementation, application security, deployment, and production operations

Chrome Privacy & Security Engineer

Google · New York, USA & Munich, Germany

Jan 2020 - Jun 2025

- Led the end-to-end remediation of a Chrome Android update-detection vulnerability that could suppress warnings for outdated, vulnerable browser versions; redesigned the client/server flow, fixed protocol and threading defects, and drove backend changes across Chrome Update infrastructure teams
- Owned Chrome Android Safety Check from technical design through launch as a prominent top-level Settings entry; obtained security, privacy, and accessibility approvals and kept leadership aligned while unifying compromised-password, Safe Browsing, and browser-update risks in one remediation workflow
- Owned 200+ end-to-end security/privacy launch reviews using data-flow tracing, trust-boundary/abuse-case analysis, architecture review, and threat modeling; drove revisions and escalated 3+ launch blocks, partnering with director-level leadership on mitigations or policy exceptions
- Selected as a Chrome telemetry approver, especially for Android; embedded reusable guidelines in the standard review workflow for 10+ teams, reducing repeat policy violations involving PII or linkable data and accelerating reviews
- Served as Chromium code owner and team mobile-development point of contact, conducting 500+ C++/Java/JNI reviews across desktop, Android, and WebView; surfaced pre-merge risks in lifecycle handling, feature gating, API boundaries, and test coverage, and mentored an engineer to independent ownership
- Blocked a bookmark launch risking exposure of sensitive data on shared devices; co-designed a safer architecture and verified design, code, and tests before approval
- Analyzed Chrome Android and CI logs to debug cross-layer failures; in recurring on-call, restored builders, rolled back breaking changes, and helped owners fix and safely reland 20+ changes

Search Infrastructure & Reliability Engineer

Google · Chicago, IL, USA

Oct 2018 - Jan 2020

- Defined operational SLAs and alert thresholds, then designed, deployed, and operated Borg jobs with Monarch monitoring and alerting across 1,500+ service instances; won cross-team approval and reported system health and trade-offs to leadership
- Standardized server configuration across 4 non-production environments, reducing configuration drift and making evaluation results more reproducible
- Improved Google Search release-quality evaluation by diagnosing and eliminating flaky infrastructure and test failures, making release signals more reliable and reproducible
- Technologies:** C++, Borg (Kubernetes precursor), Colossus, Protocol Buffers, Monarch monitoring

Software Engineer

Jump Trading · Chicago, IL, USA

Jun 2016 - Jul 2018

- Built a Splunk analytics pipeline processing 100M+ trading events/day, reducing time to detect and diagnose trading anomalies from hours to minutes
- Designed and built a C++ on-disk storage system sustaining 500K+ writes/sec at sub-millisecond p99 latency over tens of terabytes of trading data, improving replay time by 2–3x
- Developed a distributed CUDA market simulator with quantitative analysts to model latency across 20+ exchanges
- Technologies:** C++, CUDA, Splunk; distributed systems, high-throughput storage, low-latency systems

Software Engineer Intern

Dropbox · San Francisco, CA, USA

May 2015 - Aug 2015

- Built and shipped a Python/React/TypeScript distributed system for team-administration actions at 100,000-user scale

Software Engineer Intern

Apple · Cupertino, CA, USA

May 2014 - Aug 2014

- Improved iOS and OS X input-method usability and accessibility in Objective-C using Xcode and platform SDKs

Education

University of Illinois at Urbana-Champaign

Champaign, IL, USA

GPA: 4.0/4.0

Master of Science in Computer Science – Machine Learning

Jan 2017 - Dec 2017

Bachelor of Science in Computer Science

Aug 2013 - Dec 2016

Bronze Tablet Honoree, James Scholar, Dean's List

Skills

Product & Application Security: Threat modelling, secure architecture/code review, vulnerability remediation, secure SDLC, privacy/GDPR

Security Testing: Hands-on web/API testing, OWASP Top 10, Burp Suite, SSRF, access control, and API vulnerabilities

Engineering & DevSecOps: C++, Java, Python, JavaScript/TypeScript, CI/CD, containers, monitoring/alerting, Splunk, Git, Linux

Languages: English (fluent), Russian (fluent), Spanish (fluent), Portuguese (fluent), German (conversational)